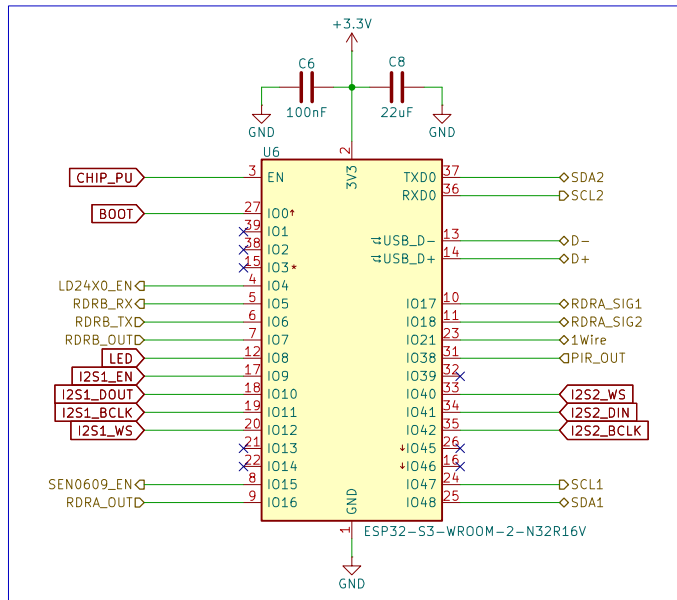
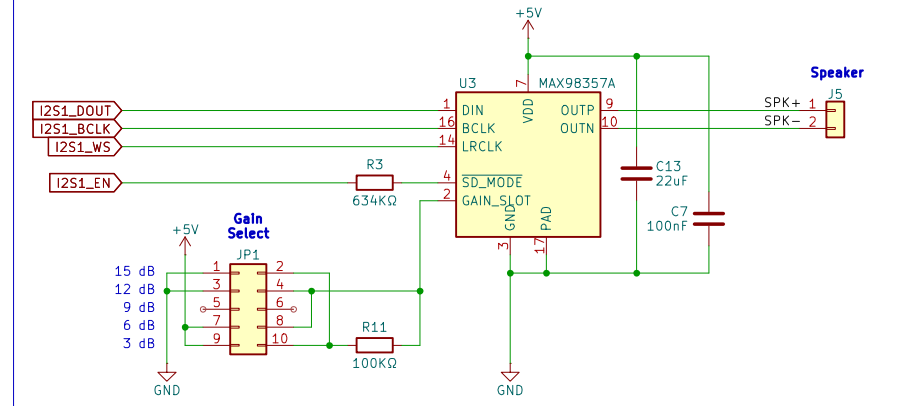


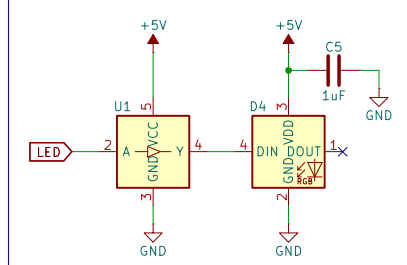
ESP32-S3



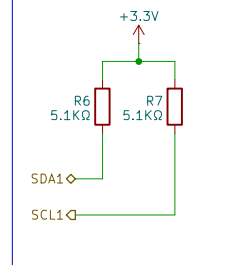
Speaker DAC and Amplifier



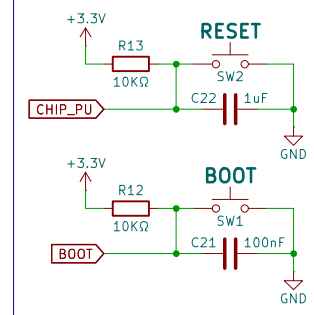
LED Status



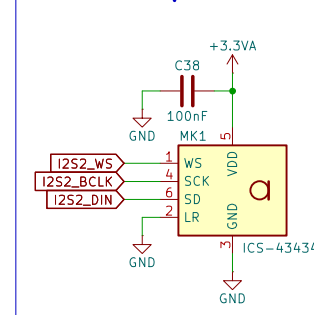
I2C1 Pullup



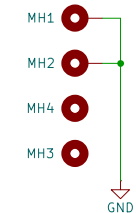
ESP Control



Microphone



Speaker Mounting Holes



Notes

- ESP32-S3 Strapping Pins: GPIO0, GPIO3, GPIO45, GPIO46.
- MAX98357A DAC when enabled is configured for stereo data mode.
- SEN0609_EN and LD24X0_EN are no longer used. In order to maintain backwards compatibility these pins should not be used.

Rev B: Support for LD2410S/LD2420 and temp sensor for CO sensor.

Rev A: Fixed Rev - mistakes.

Mike Lawrence

Sheet: /ESP32-S3/

File: ESP32.kicad_sch

Title: ESPHome Parking Assistant

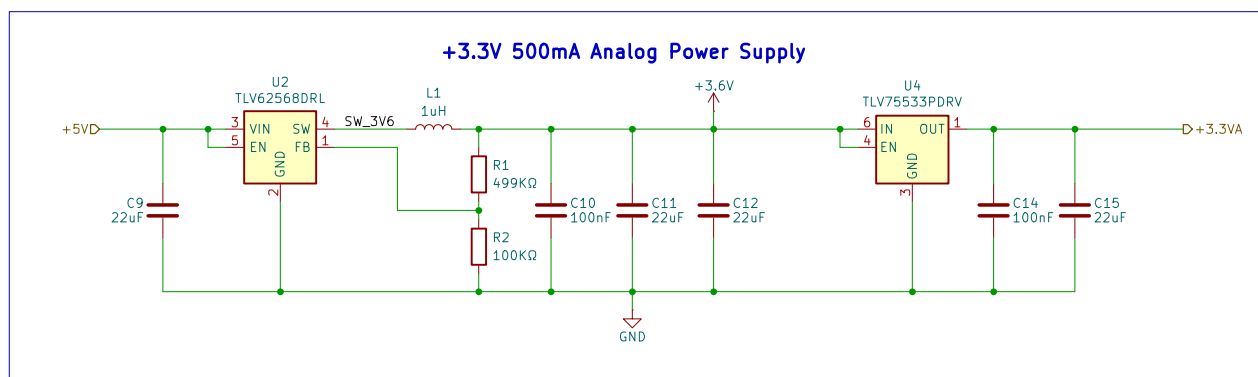
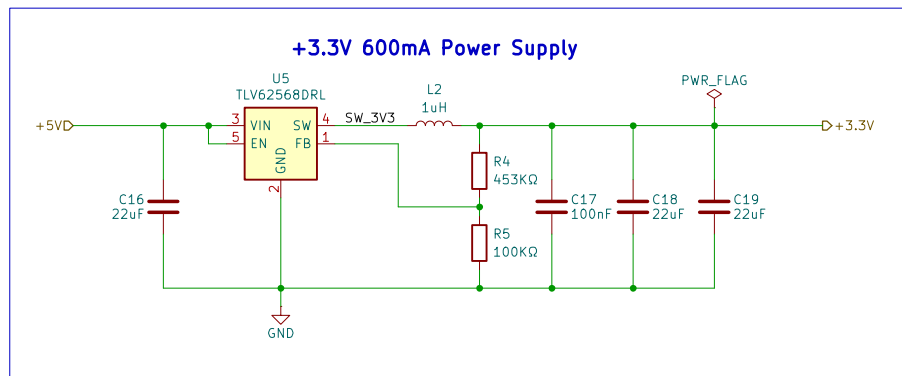
Size: A4

Date: 2025-09-23

Rev: B

KiCad E.D.A. 9.0.5-rc1

Id: 2/6



Mike Lawrence

Sheet: /Power Supplies/
File: Power-Supplies.kicad_sch

Title: ESPHome Parking Assistant

Size: A4 Date: 2025-09-23

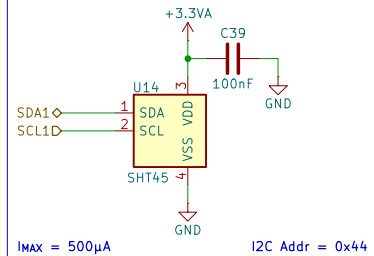
KiCad E.D.A. 9.0.5-rc1

Rev: B

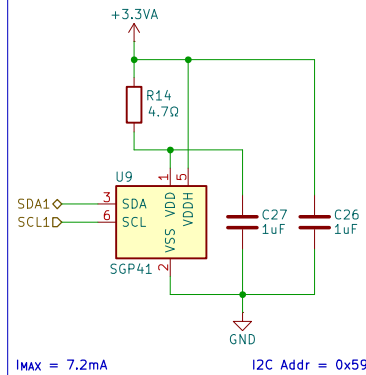
Id: 3/6

Bath or Kitchen Set

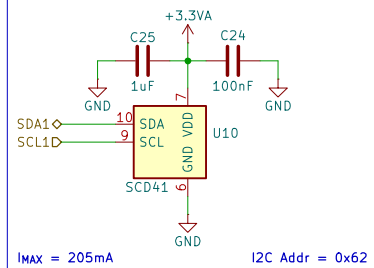
Temperature Humidity Sensor



Air Quality Sensor

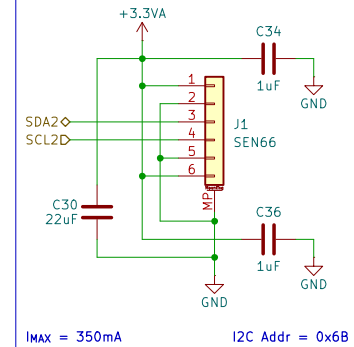


CO2 Sensor

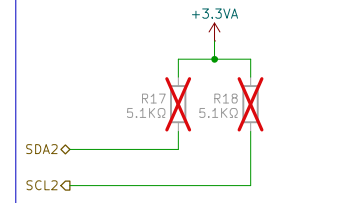


All-In-One Sensor

Environmental Sensor



I2C2 Pullup



Notes

1. Recommend one of two configurations.
 - A. SEN66.
 - B. SHT45, SCD41 and SGP41 for Bath or Kitchen.
2. Temperature & Humidity Sensor can be SHT40, SHT41, or SHT45 for increasing accuracy.
3. CO₂ (SCD40 or SCD41) measures CO₂, Temperature and Humidity.
4. Air Quality Sensor (SGP40 or SGP41) measures VOC, NO_x. These sensors are self heating and not recommended for measuring room temperature.
5. SEN66 measures Temperature, Humidity, CO₂, VOC, NO_x and PM.

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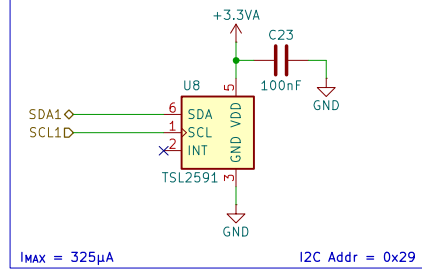
Sheet: /Environmental Sensors 1/
File: Env-Sensors-1.kicad_sch

Title: ESPHome Parking Assistant

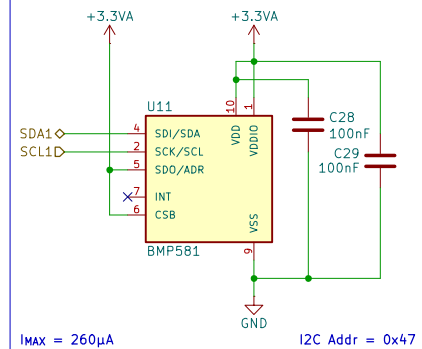
Size: A4 Date: 2025-09-23
KiCad E.D.A. 9.0.5-rc1

Rev: B
Id: 4/6

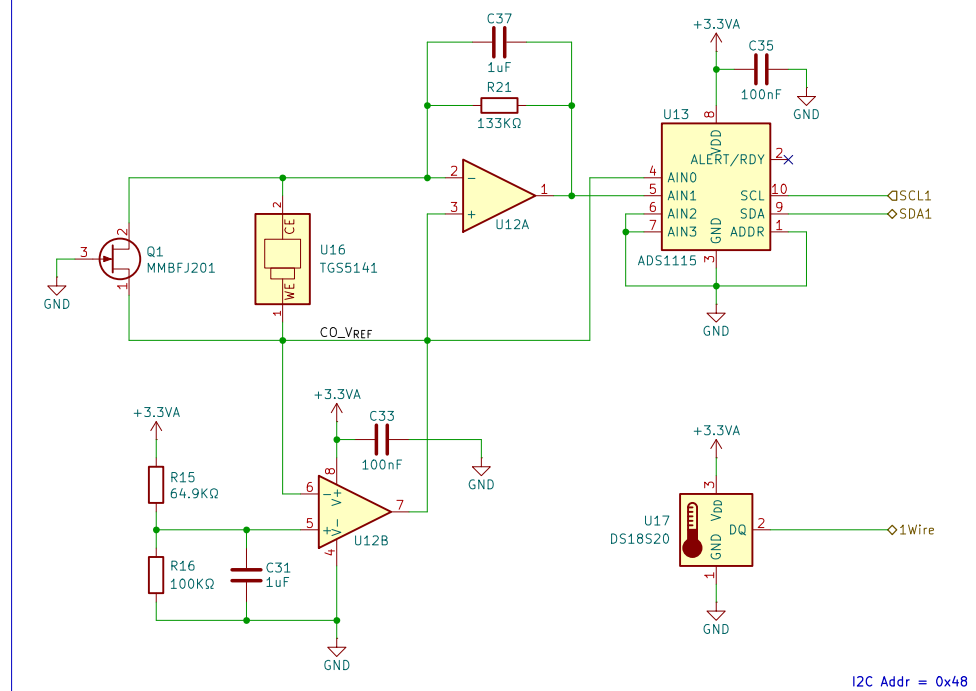
Ambient Light Sensor



Pressure Sensor



CO Sensor



Notes

1. Accuracy of CO_VREF not critical.
Needs to be around 2V to turn off Q1.
2. ADC range 0.512V.
3. Max 1000PPM.
4. Gain = $0.512V / (3.2nA \times 1.225 \times 1000PPM) = 133.333k$

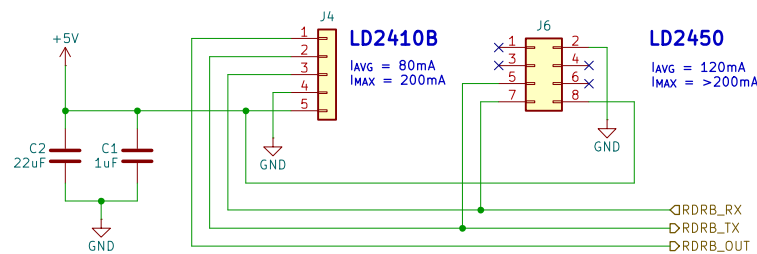
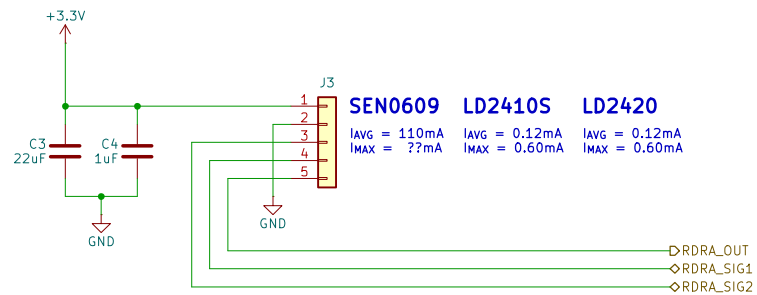
Mike Lawrence

Sheet: /Environmental Sensors 2/
File: Env-Sensors-2.kicad_sch

Title: ESPHome Parking Assistant

Size: A4 Date: 2025-09-23
KiCad E.D.A. 9.0.5-rc1

Rev: B
Id: 5/6



Notes

1. RDRA_SIG1 is SEN0609 TX, and LD2410S/LD2420 RX.
2. RDRA_SIG2 is SEN0609 RX, and LD2410S/LD2420 TX.

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Sheet: /Presence Sensors/
 File: Pres-Sensors.kicad_sch

Title: ESPHome Parking Assistant

Size: A4 Date: 2025-09-23
 KiCad E.D.A. 9.0.5-rc1

Rev: B
 Id: 6/6